# column-hse-ltau: hydrostatic columns on an optical-depth grid

Physics of `run.mode = "column-hse-ltau"` (HSEMHS). 24 Oct 2026.

## 1. What the mode does

Stokes inversions return the atmosphere on surfaces of constant optical depth, $\log_{10}\tau_{500}$ = const, not at fixed geometric heights. For such an atmosphere, $T(x, y, \tau)$ and $\mathbf{B}(x, y, \tau)$, this mode computes in every column the gas pressure $P(\tau)$, the density $\rho(\tau)$ and the **geometric height** $z(\tau)$ that hydrostatic equilibrium implies. As in `column-hse-z`, each column is independent; the only coupling between columns is the choice of their vertical offsets (Sect. 5). It is the starting state of `mhs-poisson-ltau`.

## 2. Quantities

| Symbol | Meaning | Unit |
|---|---|---|
| $\tau$, $\ln\tau$ | optical depth at 500 nm; the grid is uniform in $\log_{10}\tau$ (production: $-8 \ldots +1.4$, 0.1 dex) | – |
| $T$, $\mathbf{B}$ | temperature and magnetic field on the $\tau$ levels, from the input, fixed | K, T |
| $P$, $\rho$ | gas pressure and density, $\rho = \rho(T, P)$ (EOS) | Pa, $\mathrm{kg\,m^{-3}}$ |
| $\chi$ | extinction coefficient at 500 nm, $\chi = \chi(T, P)$ (EOS table) | $\mathrm{m^{-1}}$ |
| $\kappa$ | extinction per unit mass, $\chi/\rho$ | $\mathrm{m^2\,kg^{-1}}$ |
| $z(\tau)$ | geometric height of a $\tau$ level in a column | m |
| $g$ | gravity, 274 | $\mathrm{m\,s^{-2}}$ |

## 3. The equations

Hydrostatic equilibrium $dP/dz = -\rho g$ and the definition of optical depth $d\tau = -\chi\,dz$ combine into equations along $\ln\tau$:

$$\boxed{\frac{d\ln P}{d\ln\tau} = \frac{g\,\tau\,\rho}{\chi\,P} = \frac{g\,\tau}{\kappa\,P}, \qquad \frac{dz}{d\ln\tau} = -\frac{\tau}{\chi}}$$

The first is the classical stellar-atmosphere form $dP/d\tau = g/\kappa$: the pressure at a given optical depth is the weight of the column above it, $P \approx g\,\tau/\bar{\kappa}$. The second turns the optical-depth scale into a height scale. Both right-hand sides depend on $P$ through the EOS, so they are integrated together, down from the top level.

**Top boundary.** Above the top level $\kappa$ is taken as constant, so $P_{\rm top} = g\,\tau_{\rm top}/\kappa(T_{\rm top}, P_{\rm top})$, solved as a fixed point in $P_{\rm top}$. Since the top level is at $\tau = 10^{-8}$, an error in this assumption changes $P$ only near the top: it decays as $\tau$ grows and the integrated weight dominates.

**No anchor is needed.** Unlike the height grid, the pressure scale is fixed by $\tau$ itself (the mass above a given $\tau$), so no reference pressure is supplied. What $\tau$ does *not* fix is the absolute height of a column: $z$ is known only up to a constant per column (Sect. 5).

## 4. Between grid levels: hot gas over cool gas

Inside each 0.1 dex interval $\ln T$ is taken linear in $\log_{10}\tau$, and the equations are integrated with adaptive second-order (Heun) substeps that change $\ln P$ by at most 0.05. Where a hot layer lies directly on top of a much cooler one (for example 8200 K over 4200 K within 0.1 dex), the opacity of the cool gas is orders of magnitude lower. The linear interpolation then puts a whole 0.1 dex of nearly transparent gas below the hot layer, and $dP/d\tau = g/\kappa$ makes the pressure jump: in about 1 % of the columns $\ln P$ would rise by ∼20 within one interval ($P \to 6 \times 10^8$ Pa) and push $\tau = 1$ a megametre down. This is an artefact of representing a sharp temperature jump on a coarse $\tau$ grid, not physics. The rise of $\ln P$ is therefore capped at `cap_per_dex` = 50 per dex (5 per 0.1 dex), which still allows the physical rises (up to 5 per 0.1 dex in 99 % of the columns). Below a capped interval the pressure returns to $g\tau/\kappa$ within a few levels. On the production cube 2585 of the $\sim 2.4 \times 10^7$ intervals are capped.

## 5. Aligning the columns

Each column's height scale has an arbitrary zero. The mode chooses the offsets so that the **total pressure** $P + B^2/2\mu_0$ is equal in all columns at one height: the global median of the total pressure on the bottom $\tau$ level. The idea is lateral pressure balance at depth: deep down, flux tubes and their surroundings must push equally hard on each other, so a strong-field column, whose magnetic pressure carries part of the load, has a lower gas pressure at the same height and sits deeper (Wilson depression). Where a column's bottom total pressure is below the reference, it is extrapolated hydrostatically below its last level (8192 of 16384 columns in a test crop; by construction about

half). With `align = "none"` every column keeps $z = 0$ at $\tau = 1$.

Finally all heights are shifted together so that the mean height of $\tau = 1$ is zero (a global gauge; it has no physical content).

## 6. What the result means

- Every column is hydrostatic in $\tau$. The pressure on each $\tau$ level is essentially the weight of the gas above, independent of the neighbours. Horizontal force balance is not imposed.
- The heights come from the opacity: where the gas is opaque, a $\Delta \log \tau$ step is geometrically thin, where it is transparent the step is thick. Hot granules (high $H^-$ opacity) therefore have their $\tau = 1$ surface high, cool lanes and strong-field columns (evacuated by magnetic pressure) low.
- Production cube (`IO/input-mhs-ltau.nc`, $504 \times 504 \times 95$ levels): $P(\tau = 1)$ median 13.2 kPa; $z(\tau = 1)$ percentiles 1/50/99 $= -91/-41/+293$ km before the gauge; the $+293$ km tail is hot granules, strong-field columns are depressed. Levels near $\tau = 10^{-8}$ lie at tens of Mm in the hot columns, as hydrostatics requires.
- The magnetic field enters only through the alignment (magnetic pressure). Its tension and the horizontal force balance between columns are the subject of `mhs-poisson-ltau`.